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# Artificial Intelligence in Citizenship and Civic Education: A PRISMA 2020 Systematic Review of Digital Citizenship, AI Literacy and Democratic Engagement

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***Authors' contributions***

*This work was carried out in collaboration between both authors. Both authors read and approved the final manuscript.*

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## Abstract

**Background:** Artificial intelligence (AI) is moving quickly from a backstage administrative tool to a force that shapes how young people learn to be citizens. Yet the evidence on AI within democratic, citizenship and civic education has not been gathered systematically.
**Objective:** This review maps the peer-reviewed evidence on AI in citizenship-related education published between January 2018 and April 2026.
**Methods:** Following the PRISMA 2020 statement, structured searches across seven databases (Scopus, Web of Science, ERIC, DOAJ, Dimensions, Semantic Scholar and Crossref) returned 1,272 records; after de-duplication and two-stage screening, 27 studies met the criteria, were charted on a common extraction matrix, and were analysed using thematic synthesis.

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**Results:** A central empirical finding is the complete absence of eligible studies between 2018 and 2023: no included work predates 2024, and activity surges sharply in 2025, mapping almost exactly onto the public release of generative AI tools in late 2022. The corpus is led by China, Indonesia and the United States, and is dominated by reviews, conceptual frameworks and quantitative survey work, with only one controlled experiment. Eight themes were synthesised, the largest being governance and ethical risk, AI-literacy frameworks, and digital-citizenship formation.
**Conclusions:** AI is most often treated as something citizens must be made literate about and protected from, rather than a settled instructional method for civic learning. The field needs stronger causal designs, wider geographic coverage, and clearer links between AI literacy and substantive democratic competence.

*Keywords: Artificial intelligence; citizenship education; civic education; digital citizenship; AI literacy; misinformation; democratic engagement; PRISMA.*

## 1. Introduction

Citizenship education has always borrowed the communication tools of its age, from the printed civics primer to the classroom television and, more recently, the learning-management system. What it means to educate a "good citizen" is itself contested, ranging from the personally responsible to the participatory and justice-oriented citizen (Westheimer & Kahne, 2004), and each conception places different demands on the tools used to teach it. Artificial intelligence is the newest of these tools, but it is unusual in two respects. It does not merely deliver civic content; it can also generate it, personalise it, and in some settings moderate the very debates that civic learning depends upon. At the same time, the same technology underwrites synthetic media and algorithmic curation that can distort the information environment in which citizens form judgments (Roozenbeek & van der Linden, 2019). AI therefore sits on both sides of the civic ledger at once, as an instrument of education and as a problem that education must address.

Underlying much of this is the concept of AI literacy, defined as the competencies that let people critically evaluate, interact with, and understand AI (Long & Magerko, 2020; Ng et al., 2021). International policy has moved quickly to frame AI literacy and oversight as priorities for education systems (UNESCO, 2021), often ahead of the evidence base that would justify specific classroom practices. Some authors treat AI as a means of strengthening civic learning; others treat it primarily as a risk that civic and media-literacy education must inoculate young people against; and a third strand is concerned less with classrooms than with governance. These conversations rarely speak to one another, and a systematic review is therefore timely, not to declare the matter settled, but to establish what is actually known and where the gaps lie.

This paper offers such a synthesis for the period January 2018 to April 2026. It applies the PRISMA 2020 framework to identify, screen and chart peer-reviewed studies that connect AI with democratic, citizenship, civic or digital-citizenship education, and it analyses the resulting corpus thematically. The review is deliberately cautious in its claims: the field is young, and the included evidence is more conceptual than experimental. The contribution is a clear map of the terrain, an honest account of its limits, and a set of researchable questions for the next phase of work.

## 2. Review Rationale

Three considerations motivate this review. First, existing reviews of AI in education are broad and rarely isolate the civic dimension; syntheses tend to centre on tutoring, assessment and learning analytics (Garzón et al., 2025), with citizenship appearing only as a passing reference. Second, the policy environment has moved faster than the evidence (UNESCO, 2021), so mapping the evidence helps separate what is demonstrated from what is aspired to. Third, the topic is interdisciplinary by nature, drawing on education, political science, media studies and computer science, and an integrative review can connect findings that would otherwise remain siloed within each field.

## 3. Objectives of the Review

The review was guided by four questions:

- **RQ1.** What is the volume and temporal trend of peer-reviewed research on AI in citizenship-related education between 2018 and 2026?

- **RQ2.** Where is this research produced, and what research designs and AI technologies does it use?
- **RQ3.** What themes emerge when the studies are synthesised, and how do they relate to one another?
- **RQ4.** What gaps and implications follow for education research, practice and policy?

## 4. Methodology

### 4.1 Protocol and Reporting Framework

The review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement and its explanation and elaboration (Page et al., 2021; Page et al., 2021), which updates the original PRISMA guidance (Moher et al., 2009). The process moved through the four standard stages of identification, screening, eligibility and inclusion, summarised in Fig. 1. No meta-analysis was attempted because the studies were too heterogeneous in design and outcome to pool statistically; the synthesis is therefore narrative and thematic (Thomas & Harden, 2008).

### 4.2 Databases and Search Strategy

Searches were run across Scopus, Web of Science, ERIC, DOAJ, Dimensions, Semantic Scholar and Crossref, with Google Scholar and backward citation searching used to catch records the databases missed. The search combined three keyword clusters using Boolean logic. The core string, adapted to each database's syntax, was: *("artificial intelligence" OR "AI" OR "generative AI" OR "machine learning" OR "chatbot" OR "ChatGPT" OR "large language model") AND ("citizenship education" OR "civic education" OR "democratic education" OR "digital citizenship" OR "political literacy" OR "civic engagement" OR "media literacy" OR "misinformation") AND ("student*" OR "teacher education" OR "school" OR "higher education" OR "curriculum")*

Truncation and phrase searching balanced recall and precision. Searches were limited to the title, abstract and keyword fields where the database allowed. All searches were executed on 30 April 2026. To meet PRISMA 2020 transparency expectations, the records returned by each database are reported separately rather than as a single aggregate (Table 1); database searching returned 1,272 records, to which 18 records located through citation and hand searching were added, giving 1,290 records identified in total.

### 4.3 Eligibility Criteria

#### 4.3.1 Inclusion

Peer-reviewed journal articles or indexed proceedings; published January 2018 to 30 April 2026; written in English; carrying a traceable DOI or stable identifier; and substantively connecting AI with democratic, citizenship, civic, digital-citizenship, political-literacy, AI-ethics or misinformation education. Empirical, conceptual, mixed-method, review and policy studies were all eligible.

#### 4.3.2 Exclusion

Non-English records; conference abstracts without a full text; duplicates; items without a traceable identifier or stable URL; sources from journals flagged as predatory; general AI studies with no educational framing; and non-scholarly material such as blogs, magazine pieces and institutional press releases.

### 4.4 De-duplication, Screening and Selection

The 1,290 identified records were imported into a reference manager (Zotero 6), and duplicates were removed in two passes: an automated match on DOI and on normalised title-plus-first-author, followed by a manual check of near-duplicates flagged by fuzzy title matching. This removed 348 duplicate records, leaving 942 unique records for screening.

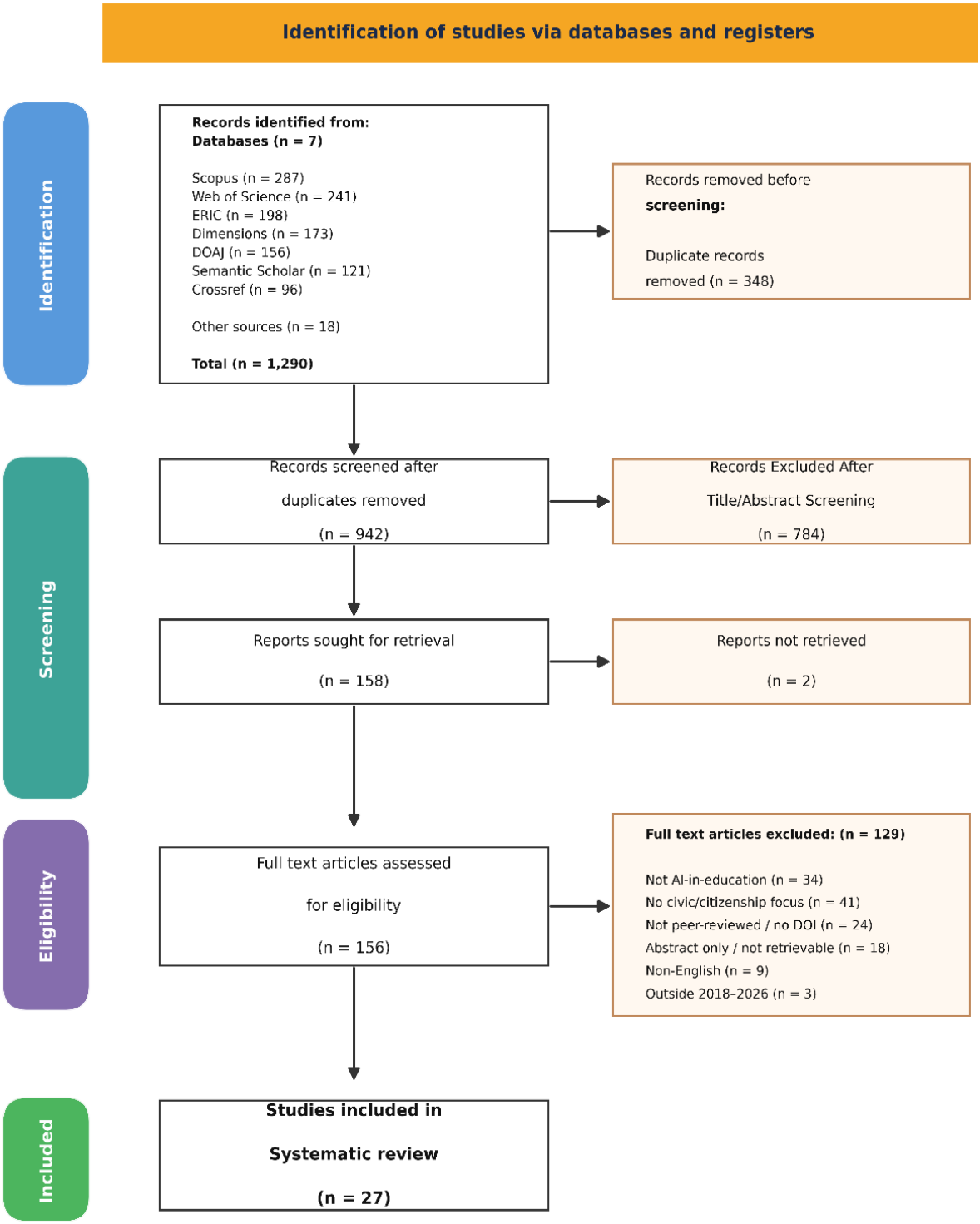



**Fig. 1. PRISMA 2020 study-selection flow diagram**

Two reviewers independently screened all 942 records on title and abstract against the eligibility criteria; disagreements were resolved by discussion, and inter-rater agreement at this stage was substantial (Cohen's $\kappa$ = .81). Screening excluded 786 records, leaving 156 reports whose full texts were assessed for eligibility. Full-text exclusions (n = 129) were logged with reasons (Fig. 1). To reduce selection bias, the eligibility criteria were fixed before screening, both reviewers applied them independently, and borderline cases on the civic-relevance criterion were retained to the full-text stage rather than excluded on the abstract alone. Twenty-seven studies

were retained for synthesis and charted on a common extraction matrix capturing author, year, country, design and sample, AI technology, citizenship focus, key findings and DOI (Table 2).

**Table 1. Records retrieved per database, with fields searched and filters applied**

| Database | Records (n) | Fields searched / filters |
|---|---|---|
| Scopus | 287 | Title-Abstract-Keyword; document type: article/review/conference paper |
| Web of Science (Core Collection) | 241 | Topic (TS) field; SCI-E, SSCI, ESCI, CPCI indexes |
| ERIC | 198 | Keyword (all fields); peer-reviewed limiter applied |
| DOAJ | 156 | Title and abstract; open-access journals only |
| Dimensions | 173 | Full-text + title-abstract; publication type: article/proceedings |
| Semantic Scholar | 121 | Title-abstract relevance ranking; top records screened to saturation |
| Crossref | 96 | Bibliographic metadata query; journal-article type |
| Total (database searching) | 1,272 | Plus 18 records from citation/hand searching = 1,290 records identified |

### 4.5 Quality Appraisal and Synthesis

Because the corpus mixed empirical, conceptual and review designs, methodological quality was appraised with design-appropriate tools: The Mixed Methods Appraisal Tool (MMAT) for empirical and mixed studies, and a transparency-of-argument check (stated aims, evidence base, and acknowledged limitations) for conceptual and review papers. Appraisal informed the weight given to each study in the synthesis rather than serving as an exclusion gate, given the field's early stage. Findings were combined using thematic synthesis (Thomas & Harden, 2008): codes were derived inductively from each study's findings, grouped into descriptive themes, and then organised into the eight analytical themes reported in Section 6.

## 5. Results

### 5.1 Volume and Temporal Trend (RQ1)

The 27 included studies are striking for their recency. Not one predates 2024 within the criteria applied here, and the bulk appeared in 2025, with early-2026 issues already adding more (Fig. 2). The earliest included works are conceptual pieces on generative AI in political education (Xu, 2024; Ran & Zeng, 2024) and the political-science survey of Wu and Wu (2024). The surge reflects the arrival of widely accessible generative AI from late 2022, which turned a niche concern into a mainstream educational question almost overnight. The practical consequence is that the field has very little longitudinal evidence, simply because there has not yet been time to gather it.

### 5.2 Geography, Design and Technology (RQ2)

Geographically, the corpus is led by China, Indonesia and the United States, which together account for more than half of the studies, followed by Germany, Hong Kong SAR and Türkiye, and a tail of single-study countries including Canada (Sieber et al., 2025), Colombia (Garzón et al., 2025), Hungary (Altamimi & Abu Al-Haija'a, 2026), Jordan (Alkhawaja et al., 2025), Portugal (Pinho et al., 2025) and Vietnam (Al Issa & Maheshwari, 2026) (Fig. 3; country is assigned by the corresponding author's institution). The Chinese contribution clusters around ideological-political education and AI-literacy measurement (Xu, 2024; Wu, Sun, et al., 2025); the Indonesian work is distinctive for tying digital citizenship to Pancasila civic values (Sutrisno et al., 2025; Warlim et al., 2026); and the United States contributes most of the social-studies and political-science teaching literature (Heafner & Maxwell, 2025; Wu & Wu, 2024). Much of the Global South, and large parts of Europe, remain thinly represented.

By design, the field leans conceptual and synthetic rather than experimental (Fig. 4). Reviews together with conceptual or framework papers make up nearly two-thirds of the corpus, quantitative surveys form most of the

remainder, and only four studies are experimental or mixed-method with primary intervention data. The single controlled experiment in the set (Warlim et al., 2026), an Indonesian cross-faculty study of AI-enhanced civic-character teaching, is also the methodologically richest, which underlines how rare such designs still are. In terms of technology, generative AI and ChatGPT dominate (Heine & König, 2025; Wu et al., 2025), with a smaller group of studies addressing AI-literacy constructs in the abstract and only a handful using bespoke tutoring platforms.

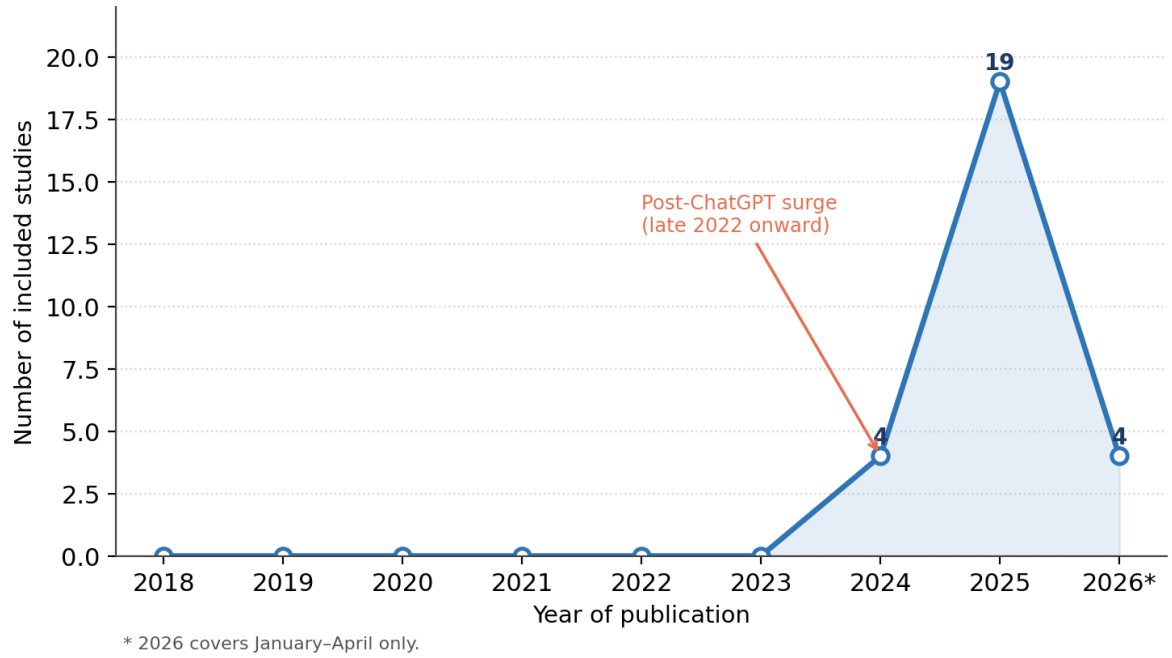



**Fig. 2. Temporal distribution of included studies**

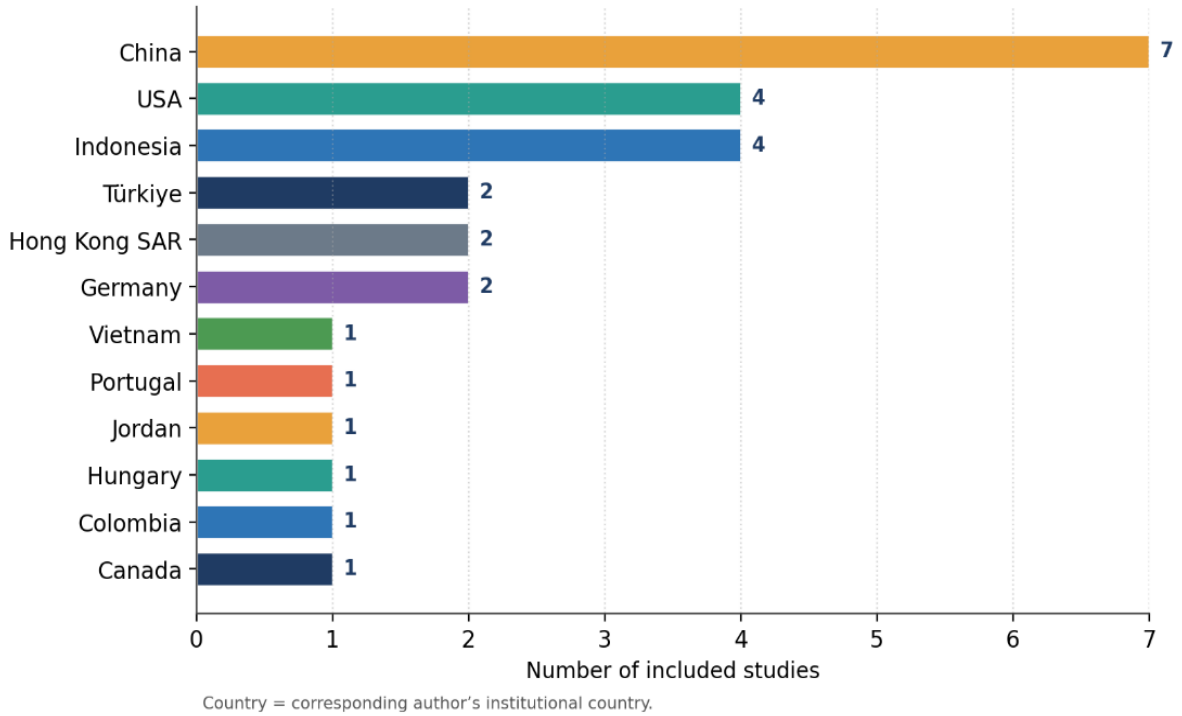



**Fig. 3. Geographical distribution of included studies**

**Table 2. Data-extraction matrix of included studies (N = 27)**

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| 1 | Altamimi, & Abu Al-Haija'a (2026) | Hungary | Conceptual / framework | AI competencies | AI literacy frameworks | Integrates AI competencies into a holistic digital-literacy framework whose pillars include digital citizenship and ethical competence. | 10.1038/s41598-026-47133-1 |
| 2 | Agus et al. (2025) | Indonesia | Mixed methods | General digital / AI tools | Digital citizenship & civic character | Digital citizenship is weakly integrated into Indonesian high-school civic education (PPKn) despite students' digital civic activity. | 10.1080/2331186X.2025.2534156 |
| 3 | Al Issa et al. (2026) | Vietnam | Bibliometric review (817 studies) | ChatGPT / generative AI | Governance, policy & ethical risk | Cluster analysis of generative-AI-in-education research; advocates AI literacy and policy reform over outright bans. | 10.1108/ITSE-01-2025-0013 |
| 4 | Alkhawaja et al. (2025) | Jordan | Quantitative | ChatGPT, Grammarly, QuillBot, etc. | AI literacy frameworks | AI tools positively influenced self-directed learning, problem-solving, critical thinking and digital literacy. | 10.3389/feduc.2025.1691148 |
| 5 | Chu-Ke et al. (2024) | Hong Kong SAR | Review / conceptual | Generative AI | Misinformation & media literacy | Argues generative AI intensifies misinformation and calls for renewed media-/AI-literacy interventions. | 10.1177/27523543241240285 |

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| 6 | Garzón et al. (2025) | Colombia | Systematic review (155 studies) | Multiple AI types | Governance, policy & ethical risk | Reviews AIED trends, benefits and challenges (2015–2025), noting a post-2022 surge and persistent ethics/equity concerns. | 10.3390/mti9080084 |
| 7 | Gu et al. (2025) | USA | Integrative review (124 studies) | Generative AI | AI literacy frameworks | Distinguishes functional from critical AI literacy; notes thin empirical evaluation in both K-12 and higher education. | 10.1145/3702652.3744217 |
| 8 | Heafner et al. (2025) | USA | Conceptual / framework | Generative AI / LLMs | Digital citizenship & civic character | Introduces the CIVIC framework (five pillars) for using AI in social-studies teaching to cultivate future-ready, critical citizens. | *CIVIC: Five Pillars for Using Artificial Intelligence in Social Studies Education – CITE Journal* |
| 9 | Heine et al. (2025) | Germany | Quantitative (survey, n=1,687) | ChatGPT | Teacher readiness | Pre-service teachers' digital self-efficacy was significantly associated with attitudes toward and use of ChatGPT. | 10.1080/02619768.2025.2540791 |
| 10 | Hill & Conceição (2025) | USA | Conceptual | Generative AI / deepfakes | Misinformation & media literacy | Frames faculty responsibility for AI-generated disinformation; offers source-evaluation and | 10.1002/ace.70018 |

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| | | | | | | verification strategies. | |
| 11 | Khaz et al. (2025) | Indonesia | Qualitative (desk / normative) | AI in political system | Civic participation & democratic engagement | Analyses AI use in Indonesia's political system and its implications for democratic literacy and civic learning. | *https://journal.civiliza.org/index.php/gej/article/view/1255* . |
| 12 | Ma et al. (2025) | Hong Kong SAR | Systematic review (68 studies) | AI ethics education | AI ethics literacy | Maps K-12 AI-ethics education, finding east–west disparities and gaps in assessing affective/behavioural outcomes. | 10.1016/j.caeai.2025.100422 |
| 13 | Özüdoğru et al. (2025) | Türkiye | Quantitative (SEM) | General AI / AI literacy | Teacher readiness | Pre-service teachers' AI readiness was predicted by AI literacy, digital citizenship and AI-enhanced innovation. | 10.1177/02666669251335657 |
| 14 | Pinho et al. (2025) | Portugal | Scoping review | Generative AI | Governance, policy & ethical risk | Proposes a living generative-AI governance model for educational research at macro, meso and micro levels. | 10.3389/feduc.2025.1594343 |
| 15 | Ran et al. (2024) | China | Conceptual | Generative AI (ChatGPT) | Political / ideological education | Proposes scenario-based use of generative AI in ideological-political education while warning of ideological-control risks. | 10.1145/3696500.3696575 |

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| 16 | Sieber et al. (2025) | Canada | Multidisciplinary review | AI / NLP for participation | Civic participation & democratic engagement | Synthesises pathways and barriers for civic participation in AI governance across multiple disciplines. | 10.1177/23998083241296200 |
| 17 | Solak Berigel et al. (2026) | Türkiye | Qualitative (interviews, n=60) | AI in teaching | Digital citizenship & civic character | Student teachers framed digital citizenship as ethical, safe, responsible online conduct; flagged weak ethics/privacy teaching. | https://iojet.org/index.php/IOJET/article/view/2310 . |
| 18 | Stracke et al. (2025) | Germany | Conceptual / policy | AIED (broad) | Governance, policy & ethical risk | Argues education is not a standard AI domain and calls for international regulation to protect human rights, democracy and equity. | 10.1007/978-3-031-99267-4_37 |
| 19 | Sutrisno et al. (2025) | Indonesia | Quantitative (survey) | AI-based project learning | Teacher readiness | Students in a Pancasila & citizenship-education programme showed high readiness for AI-based project learning for local-wisdom values. | 10.1080/2331186X.2025.2582948 |
| 20 | Tan et al. (2025) | China | Systematic review (22 studies) | AI literacy | AI literacy frameworks | Reviews K-12 AI-literacy research, linking it explicitly to digital citizenship; | 10.1080/10494820.2025.2482586 |

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| | | | | | | most studies mixed-method and secondary-level. | |
| 21 | Warlim et al. (2026) | Indonesia | Mixed methods (2×2 experiment, n=240) | Custom AI tutoring platform | Digital citizenship & civic character | AI-enhanced civic-character teaching raised digital-citizenship (d=0.73) and civic-character (d=0.51) scores; digital citizenship mediated ~47% of the effect. | 10.3389/feduc.2025.1690799 |
| 22 | Wu et al. (2025) | China | Quantitative (scale construction, n=1,651) | AI literacy (KSAVE model) | AI literacy frameworks | Builds and tests a multi-level AI-literacy evaluation system for university students embedding AI-ethics dimensions. | 10.1007/s44366-025-0039-x |
| 23 | Wu et al. (2025) | China | Systematic review (99 studies) | Generative AI | Governance, policy & ethical risk | Synthesises responses, attitudes and use of generative AI in higher-education classrooms, including ethical self-efficacy. | 10.3390/bs15040467 |
| 24 | Wu et al. (2024) | USA | Empirical survey | Generative AI / LLMs | Political / ideological education | Surveys generative AI's impact on political-science teaching, assessment integrity and undergraduates' civic skills. | 10.1017/S1049096524000167 |
| 25 | Xu (2024) | China | Conceptual | ChatGPT | Political / | Maps opportunities | 10.54097/7rhpqh41 |

| # | Author(s) (Year) | Country | Design / sample | AI tool / technology | Citizenship focus | Key finding | DOI |
|---|---|---|---|---|---|---|---|
| | | | | | ideological education | (personalisation, engagement) and challenges (value bias, dependency) of ChatGPT for ideological-political education. | |
| 26 | Yang et al. (2025) | China | Quantitative (PLS-SEM, n=482) | Generative AI tools | AI ethics literacy | A validated AI-ethics-literacy framework (fairness, privacy, human-centricity, accountability) shaped students' learning competence. | 10.1038/s41598-025-21977-5 |
| 27 | Zhu et al. (2025) | China | Systematic review + grounded theory | AIED systems | Governance, policy & ethical risk | Classifies AIED ethical risks across technology, education and society dimensions, including citizenship and accountability risks. | 10.1057/s41599-025-05252-6 |

*Author column shows the first author for space; full author lists appear in the references. Country = corresponding author's institutional country. Open-access items without an assigned DOI are marked accordingly and cited by stable URL in the references*

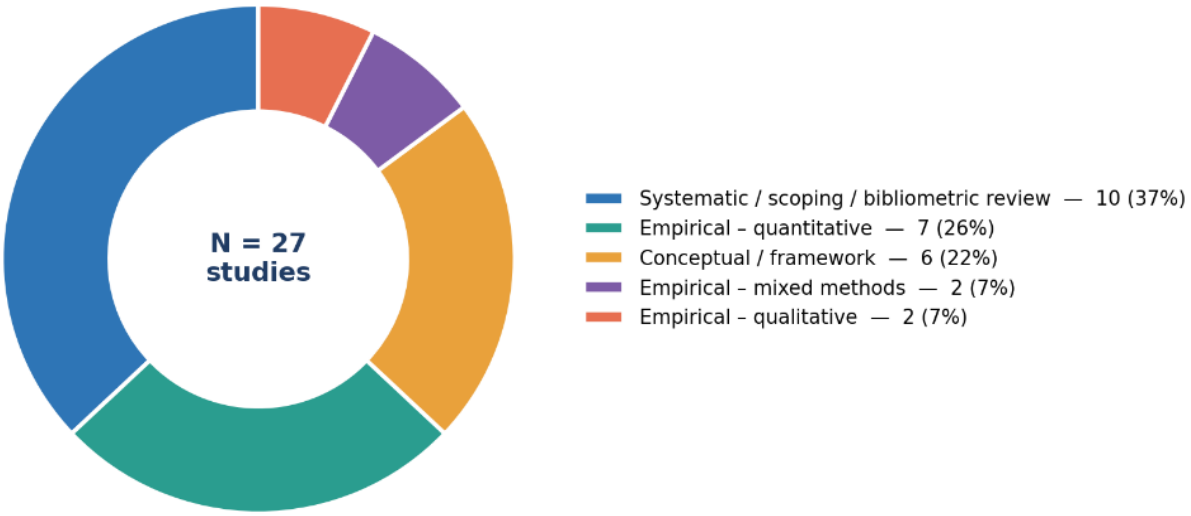



**Fig. 4. Distribution of research designs**

## 6. Thematic Analysis (RQ3)

Charting the studies by their primary concern produced eight themes (Fig. 5; Table 3 maps each theme to its research questions and contributing studies). They are presented in descending order of weight, but the more important point is how they interlock: literacy themes feed civic-formation themes, which in turn raise the governance questions that the largest theme addresses.

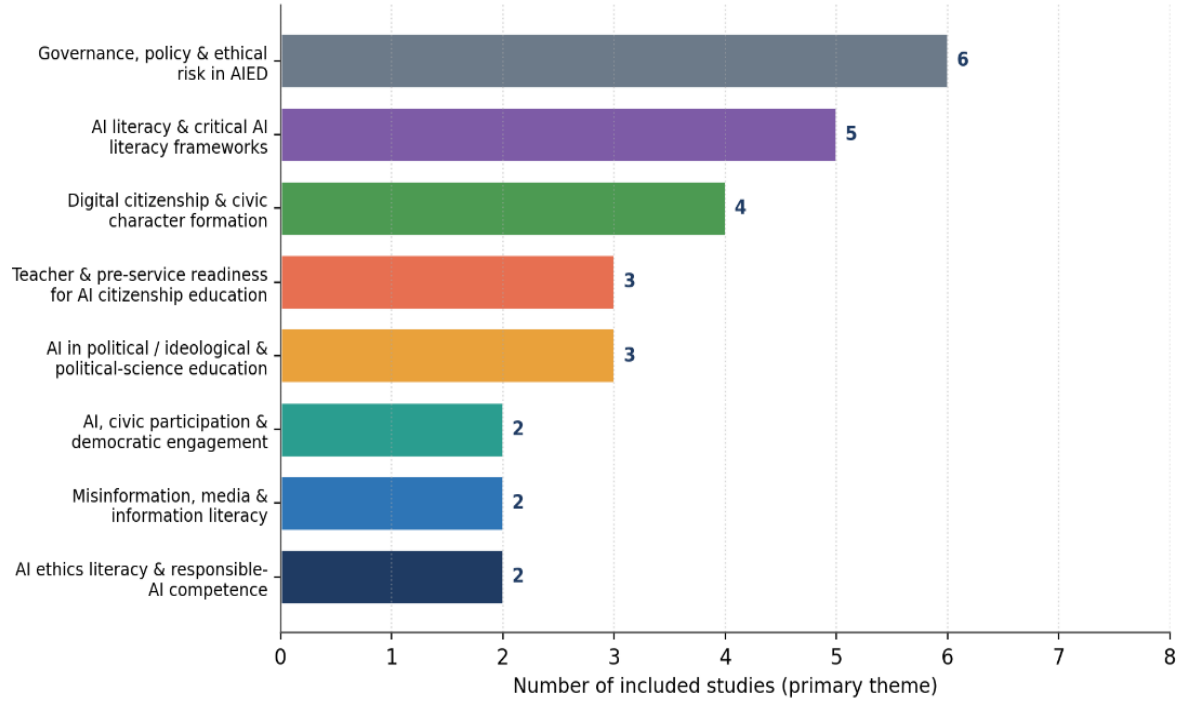



**Fig. 5. Thematic classification of included studies**

### 6.1 Governance, Policy and Ethical Risk

The largest theme treats AI less as a teaching aid than as a governance problem for education systems. Studies here classify the ethical risks of AI in education across technological, educational and societal dimensions (Zhu

et al., 2025) and argue that because education shapes civic values and reaches minors, it cannot be regulated like any other AI application (Stracke et al., 2025). Governance models have been proposed for educational research (Pinho et al., 2025), and large reviews confirm that ethics, equity and dependency concerns persist across the wider AIED literature (Garzón et al., 2025; Wu et al., 2025; Al Issa & Maheshwari, 2026). This theme gives the field its critical backbone, but it is also the most conceptual, with little empirical testing of whether particular governance arrangements actually protect civic learning.

**Table 3. Synthesis matrix mapping themes to research questions and contributing studies**

| Theme | k | Linked RQs | Contributing studies (first author, year) |
|---|---|---|---|
| Governance, policy & ethical risk | 6 | RQ3, RQ4 | Zhu et al. (2025) ; Stracke et al. (2025) ; Pinho et al. (2025) ; Garzón et al. (2025); Wu et al. (2025); Al Issa & Maheshwari (2026) |
| AI literacy & critical AI-literacy frameworks | 5 | RQ2, RQ3 | Tan & Tang (2025) ; Gu & Ericson (2025) ; Wu et al. (2025); Alkhawaja et al. (2025); Altamimi & Abu Al-Haija'a (2026) |
| Digital citizenship & civic-character formation | 4 | RQ3 | Warlim et al. (2026) ; Agus et al. (2025); Heafner & Maxwell (2025); Solak Berigel et al. (2026) |
| AI in political / ideological education | 3 | RQ3 | Xu (2024); Ran & Zeng (2024); Wu & Wu (2024) |
| Teacher & pre-service readiness | 3 | RQ2, RQ4 | Özüdoğru & Yıldız Durak (2025) ; Sutrisno et al. (2025); Heine & König (2025) |
| AI-ethics literacy & responsible-AI competence | 2 | RQ3 | Yang et al. (2025) ; Ma et al. (2025) |
| Misinformation, media & information literacy | 2 | RQ3, RQ4 | Chu-Ke & Dong (2024); Hill & Conceição (2025) |
| AI, civic participation & democratic engagement | 2 | RQ3, RQ4 | Sieber et al. (2025) ; Khaz et al. (2025) |

*Note. k = number of included studies for which the theme is primary; studies may inform additional themes secondarily*

## 6.2 AI-literacy and Critical-AI-literacy Frameworks

A second cluster develops frameworks and measurement tools for AI literacy, increasingly distinguishing functional literacy from critical literacy (Gu & Ericson, 2025), echoing the foundational competency models of the field (Long & Magerko, 2020; Ng et al., 2021). Reviews route AI literacy explicitly through digital citizenship (Tan & Tang, 2025), validated evaluation systems embed AI-ethics dimensions (Wu, Sun, et al., 2025), and applied studies link AI-tool use to critical thinking and digital literacy (Alkhawaja et al., 2025); broader frameworks fold AI competence into holistic digital literacy (Altamimi & Abu Al-Haija'a, 2026). Analytically, the weakness is one of measurement validity: most instruments capture self-reported confidence rather than demonstrated competence, few report construct validation beyond internal-consistency coefficients, and the strongest psychometric work (Yang et al., 2025; Wu et al., 2025) validates AI-literacy or AI-ethics constructs without then linking them to any behavioural civic outcome. The field therefore elaborates what AI literacy is faster than it demonstrates that measured AI literacy translates into the digital-citizenship or democratic competence it is assumed to support.

## 6.3 Digital Citizenship and Civic-character Formation

A third group studies how AI engagement relates to digital-citizenship competence and civic character. The strongest evidence is the Indonesian experimental work of Warlim et al. (2026), in which an AI-supported programme improved digital-citizenship and civic-character scores, with digital citizenship mediating much of the effect. Qualitative work describes student teachers framing digital citizenship as ethical, safe and responsible online conduct (Solak et al., 2026), while survey work finds digital citizenship weakly integrated into school civic curricula (Agus et al., 2025). Conceptually, the CIVIC framework offers five pillars for using AI in social-studies teaching (Heafner & Maxwell, 2025). The theme suggests AI can support civic-character work, but dependence on self-report and short timeframes tempers any strong causal claim.

### 6.4 AI in Political and Ideological Education

A distinct strand, concentrated in China and the United States, looks at AI within explicitly political education. Conceptual analyses weigh personalisation and engagement against value bias and dependency in ideological-political coursework (Xu, 2024; Ran & Zeng, 2024), while survey evidence examines generative AI's effect on political-science teaching and assessment integrity (Wu & Wu, 2024). Together they show that the civic uses of AI are culturally situated; what counts as desirable civic formation differs sharply across these systems.

### 6.5 Teacher and Pre-service Readiness

Because civic learning is teacher-mediated, several studies examine whether teachers are ready to use AI for it. Pre-service teachers report AI readiness shaped by AI literacy, digital citizenship and perceived AI threats (Özüdoğru & Yıldız Durak, 2025), high readiness for AI-based project learning in citizenship programmes (Sutrisno et al., 2025), and self-efficacy linked to ChatGPT use (Heine & König, 2025). The implication is consistent: technical enthusiasm runs ahead of pedagogical and ethical preparation.

### 6.6 AI-ethics Literacy and Misinformation

Two smaller themes round out the picture. AI-ethics-literacy studies model students' ethical knowledge and competence and link them to responsible AI use (Yang et al., 2025; Ma et al., 2025). Misinformation and media-literacy studies argue that generative AI has raised the stakes of the information environment and call for renewed inoculation and verification skills (Chu-Ke & Dong, 2024; Hill & Conceição, 2025), consistent with experimental evidence that inoculation can build resistance to online misinformation (Roozenbeek & van der Linden, 2019).

### 6.7 Civic Participation and Democratic Engagement

Finally, a participation-oriented theme moves beyond the classroom to ask how AI reshapes civic participation itself (Sieber et al., 2025), including AI's role in national political systems and its implications for democratic literacy (Khaz et al., 2025). This work warns that AI can introduce new participation divides, a caution directly relevant to how civic education should prepare young people for AI-mediated public life.

## 7. Discussion

Read together, the studies tell a coherent story. The field's centre of gravity is not the use of AI to teach civics, but the project of making citizens literate about, and resilient to, AI. Governance (Stracke et al., 2025; Zhu et al., 2025), literacy (Tan & Tang, 2025; Gu & Ericson, 2025) and information-evaluation themes (Chu-Ke & Dong, 2024) outweigh studies of AI as an instructional method, and even the civic-formation work tends to position digital citizenship as the thing being protected. This is a defensible response to a powerful and unsettled technology, but it means the optimistic claims often heard in policy circles (UNESCO, 2021) rest on a thin empirical base.

These tensions surface as concrete contradictions across the corpus. Where Warlim et al. (2026) report sizeable AI-driven gains in digital citizenship, (Agus et al., 2025) find digital citizenship barely integrated into the very civic curricula that are meant to deliver it a gap between experimental potential and classroom reality. On readiness, the high self-efficacy reported by pre-service teachers (Özüdoğru & Yıldız Durak, 2025; Sutrisno et al., 2025) sits awkwardly beside evidence that their preparation programmes lag behind (Heine & König, 2025), suggesting confidence may outrun competence. And on the central question of AI's civic value, the optimistic, opportunity-framed accounts of generative AI in political education (Xu, 2024; Ran & Zeng, 2024) are directly countered by the risk-framed governance and misinformation literatures (Stracke et al., 2025; Chu-Ke & Dong, 2024). Rather than a settled consensus, the field holds these positions in unresolved tension, which is consistent with its early stage.

It is worth being candid about the maturity of the field. With no included study before 2024 and only one controlled experiment, this is an emerging area rather than a consolidated one. That is itself a finding. It explains why the synthesis is necessarily thematic, and why the most useful contribution at this stage is a clear research agenda rather than a confident verdict.

## 8. Research Gaps

- Causal evidence is scarce. Beyond a single experiment (Warlim et al., 2026), the field lacks controlled or longitudinal studies linking AI use to measurable civic outcomes.
- Outcomes are mostly self-reported. Few studies use behavioural, performance-based or peer-assessed measures of civic competence.
- Geographic concentration is severe. A handful of countries dominate, and much of the Global South, Africa and South Asia in particular, is absent.
- Literacy and democracy are loosely connected. AI-literacy constructs are rarely validated against substantive democratic competence or participation.
- School-level and younger-learner evidence is thin. Higher education is over-represented relative to schools, where civic education is most formally delivered.

## 9. Implications for Education and Policy

For practice, the readiness studies point to a clear priority: teacher preparation should pair technical AI skills with the pedagogical and ethical judgment that civic teaching demands (Özüdoğru & Yıldız Durak, 2025; Heine & König, 2025). For curriculum, AI literacy is most useful for citizenship when it is taught as critical literacy (Gu & Ericson, 2025; Long & Magerko, 2020) and explicitly tied to digital-citizenship and media-literacy outcomes (Tan & Tang, 2025). For policy, individual literacy and institutional regulation are complements, not alternatives (Stracke et al., 2025; UNESCO, 2021); protecting civic learning from AI's risks requires oversight as well as education.

## 10. Future Research Directions

- Run controlled and longitudinal trials that link specific AI uses to behavioural civic outcomes, not only attitudes.
- Develop and validate measures that connect critical AI literacy to actual democratic participation and judgment.
- Deliberately commission work from under-represented regions to test whether current findings travel.
- Study school settings and younger learners, where civic education is formally situated.
- Investigate the governance question empirically: do particular oversight arrangements measurably protect civic learning?

## 11. Limitations

This review has limits that bear on how its conclusions should be read. The English-language and identifier-availability criteria will have excluded relevant work published in other languages or in venues without persistent identifiers, which may understate contributions from regions already under-represented here. Reliance on seven databases, while broad, cannot guarantee exhaustive coverage of a fast-moving and interdisciplinary field, and the relevance-ranked screening of Semantic Scholar and Crossref introduces a degree of judgment at the identification stage. The synthesis is thematic and interpretive rather than statistical, so the theme weightings reflect coded judgments about each study's primary concern, and the small corpus (N = 27) means single studies can carry disproportionate weight within a theme. Finally, because the field is so new, today's map will date quickly; this synthesis is best read as a baseline for an area that is still taking shape.

## 12. Conclusion

Artificial intelligence has entered citizenship and civic education less as a teaching method than as a civic challenge in its own right. Perhaps the clearest empirical signal in this review is temporal: the complete absence

of eligible studies before 2024 and the sharp concentration in 2025 show a field that did not exist in scholarly form until the public release of generative AI tools made it unavoidable. The evidence from 2018 to 2026 is therefore recent, geographically narrow and weighted towards governance, literacy and information-evaluation rather than demonstrated instructional benefit. The most secure conclusion is also the most modest: educators and policymakers are right to treat AI literacy and oversight as civic priorities, but they should not yet assume that AI improves civic learning, because the studies needed to show that have, for the most part, not been done. The agenda set out here is an invitation to do them.

## Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

## Competing Interests

Authors have declared that no competing interests exist.

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